Instructor Companion: Day 1 Evidence Workbooklet

Visible Output and Stored State

Companion for ‘DAY01_VISIBLE_OUTPUT_EVIDENCE_WORKBOOKLET_STUDENT_06192026_v2.pdf‘.

1 Artifact identity and use

Student artifact	Student workbooklet PDF and source in the v2 package folder.
Instructor artifact	Instructor companion/key PDF and source in the v2 package folder.
Role	Guided workbooklet, not the mastery quiz. Intended to build the evidence habit before a harder Day 1 quiz is created.
Print design	Duplex packet with twoside inner/outer margins for left-edge stapling. Page information is in the footer, not in a crowded top header.
Source boundary	Original hallway cart-test scenario. No secure exam/review prompt text, answer-key language, private student data, or near-copy source structure used.
Canvas status	Not Canvas-released. Student PDF would still need Lance upload plus Ally/Panorama scores if selected for use.

Pedagogical status

This v2 workbooklet intentionally keeps scaffolding while increasing challenge. It should be reviewed for workbook feel, page count, print comfort, and difficulty before we build the separate Day 1 quiz.

2 Point map and mastery tags

Section	Evidence collected	Pts.	Tags
1	Identify measurement variables, calculated value, display-only expression, and overwrite line.	4	recognize; predict
2	Complete state table and explain when values become 5.0 and 5.2.	8	trace; predict
3	Separate printed output from notebook display and explain Cell 2/Cell 3 evidence.	6	predict; trace; explain
4	Debug teammate lab note, write checks, and name a physical measurement condition.	10	debug; explain; test
5	Compare separate-adjustment versus immediate-overwrite edits.	7	debug; implement; explain
6	Transfer explanation and self-assessed mastery/support route.	4	transfer; explain
Total	Suggested workbook evidence score. Instructor may convert to completion/participation if preferred.	39	multi-tag workbook evidence

3 Expected state trace

Checkpoint	trial	time	speed	adjusted	Reason
After Cell 1	`cart-A`	`3.0`	`4.0`	not defined	`12.0 / 3.0`.
After Cell 2	`cart-B`	`2.4`	`5.0`	not defined	New time overwrites `time_s`; speed recalculated as `12.0 / 2.4`. Final bare expression displays `5.2` but does not store it.
After Cell 3	`cart-B`	`2.4`	`5.0`	`5.2`	Adjustment is stored separately in `adjusted_speed`; `speed_mps` remains the measured speed.
After Cell 4	`cart-B`	`2.4`	`5.2`	`5.2`	`speed_mps = adjusted_speed` overwrites the recorded speed variable.

4 Expected visible evidence

Cell	Printed output	Notebook display
Cell 1	`trial cart-A`; `speed 4.0`	none
Cell 2	`trial cart-B`	`5.2`
Cell 3	`stored 5.0`; `adjusted 5.2`	none
Cell 4	`recorded 5.2`	none

5 Prompt-level answer notes

Prompt family	Secure response	Common issue / support move
Measurement inputs	`distance_m` and `time_s`; `trial_id` labels the run.	Students may treat all variables as measurements. Ask which variables came from the hallway setup versus calculation.
First speed calculation	`speed_mps = distance_m / time_s`.	If they name a print line, return to store vs show.
Cell 2 display-only value	`speed_mps + 0.2`. It displays `5.2` because it is the final expression, but it is not assigned.	Key misconception: screen evidence equals stored value. Ask where the equals sign is.
Cell 4 overwrite	`speed_mps = adjusted_speed`.	Students may not notice that reusing the same variable name changes the current stored value.
Cell 2 explanation	`5.2` is displayed, while `speed_mps` remains `5.0`.	Ask them to run/check `print(speed_mps)` after Cell 2.
Cell 3 evidence	Printing both values separates the measured speed from the adjusted candidate.	If answer says both are 5.2, return to state trace.
Teammate note	Accurate: notebook showed `5.2`. Inaccurate: after Cell 2, `5.2` was not recorded/stored in `speed_mps`. After Cell 4, the adjusted value is recorded.	Prompt students to name exact checkpoint: after Cell 2 versus after Cell 4.
Non-mutating checks	After Cell 2: `print(speed_mps)` or final expression `speed_mps`. After Cell 3: `print(adjusted_speed)` or final expression `adjusted_speed`.	If they write an assignment, mark as mutating, not checking.
Physical measurement condition	Time must be positive and nonzero; the timed run should correspond to the measured 12-meter path.	This is a testing habit, not a syntax item. Accept plain-language statements.

6 Edit-comparison answer notes

Item	Expected answer
Version A: `speed_mps`	`5.0`; the adjustment is stored separately.
Version A: `adjusted_speed`	`5.2`.
Version B: `speed_mps`	`5.2`; the original measured speed variable has been overwritten immediately.
Better audit trail	Version A is generally better if the team wants to preserve measured speed and adjustment separately before deciding what to record. Version B may be acceptable only if the rule is already approved and immediate overwrite is intended.
Why Cell 4 is safer	It delays overwrite until after the measured speed and adjusted candidate have both been made visible. That creates a clearer lab-record trail.

7 Mastery interpretation

Band	Evidence pattern	Support route
Secure	Student traces state after each cell, distinguishes print/display/store, explains the Cell 2 trap, and can justify delayed overwrite.	Ready for harder Day 1 mastery quiz.
Developing	Student calculates most values but confuses display with storage or misses overwrite timing.	Use a two-cell recovery trace, then ask for one non-mutating check.
Needs support	Student cannot yet identify assignment versus print/display evidence.	Model a smaller example: assign, print, bare expression, overwrite. Complete first row together.

8 Optional recovery mini-surface

Use only if several students miss the Cell 2 display/storage distinction.

```
reading = 8.0
reading + 0.5
print(reading)
reading = reading + 0.5
print(reading)
```

Ask students to label what is displayed, what is printed, and what is stored after each line. Secure recovery answer: the first bare expression can display `8.5` but does not change `reading`; the assignment line later changes `reading` to `8.5`.

9 Release/accessibility notes

Local v2 workbooklet verification should check tagged PDF output, extractable text, page count under 10 for the student workbooklet, print-safe margins, and render sheets. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores. Working release threshold remains Ally at least 95% and Panorama at least 95%.